

DECORAH, IA, USA

WMO: 725476

Lat: 43.275N

Lon: 91.739W

Elev: 1157

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-4.0	-17.2	2.2	-7.6	-12.7	2.9	-4.0	22.1	14.2	19.9	17.5	4.9	310	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	89.8	75.2	86.2	73.5	83.7	72.2	77.8	85.4	75.8	83.4	74.2	81.4	7.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	137.8	82.3	73.1	128.3	80.6	72.1	124.1	79.8	42.1	85.6	40.1	83.6	38.6	81.5	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.8	15.1	DB	-15.9	92.6	6.3	3.5	-20.5	95.1	-24.2	97.2	-27.7	99.1	-32.4	101.7	(4)
(5)			WB	-16.0	80.6	5.5	2.2	-20.0	82.2	-23.2	83.4	-26.3	84.7	-30.3	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	19.6	22.5	35.1	48.2	59.5	69.0	72.7	69.4	63.0	50.7	37.3	24.2
	DBStd	21.05	12.72	12.55	12.40	10.16	8.74	6.58	5.64	5.31	8.04	9.34	10.80	11.44
	HDD50	3686	944	770	488	151	17	0	0	0	6	108	402	800
	HDD65	6993	1409	1189	930	510	217	32	7	18	132	452	833	1264
	CDD50	2851	0	1	26	96	310	570	705	601	394	128	19	1
	CDD65	685	0	0	3	5	45	153	247	154	71	7	0	0
	CDH74	5580	0	0	16	67	418	1284	2100	1085	542	66	2	0
	CDH80	1554	0	0	2	14	119	383	656	244	125	11	0	0
Wind	WSAvg	6.3	6.9	7.0	7.5	8.3	6.9	5.5	4.2	4.2	5.2	6.4	7.0	6.6
Precipitation	PrecAvg	36.9	1.1	1.0	1.9	4.0	4.5	5.8	4.9	4.8	3.1	2.4	2.1	1.3
	PrecMax	47.1	3.2	2.4	4.0	9.8	12.3	10.3	10.8	14.7	6.5	6.5	6.1	3.0
	PrecMin	24.6	0.3	0.0	0.1	1.7	1.7	2.7	1.0	1.0	1.3	0.7	0.1	0.3
	PrecStd	5.6	0.6	0.6	1.1	2.0	2.1	2.5	2.4	2.9	1.7	1.5	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	55.5	75.1	81.4	89.8	91.4	93.3	89.8	88.3	81.5	70.4	55.3
		MCWB	42.6	48.7	63.5	62.9	72.1	75.6	77.0	76.2	72.4	63.6	57.6	52.5
	2%	DB	43.0	48.1	65.7	73.4	82.2	88.1	90.0	85.8	83.6	74.8	63.3	45.8
		MCWB	38.2	44.6	59.0	58.3	66.4	73.9	76.7	74.2	71.3	61.5	54.5	42.3
	5%	DB	37.4	43.2	59.0	69.7	78.8	84.2	86.3	82.1	80.8	70.2	57.2	41.5
		MCWB	34.5	38.9	51.9	56.3	64.6	71.7	74.9	72.3	68.8	60.0	50.2	37.9
	10%	DB	35.8	38.6	52.1	64.1	73.5	81.4	83.7	80.7	76.9	65.7	53.6	37.3
		MCWB	33.8	35.1	46.4	53.1	62.2	69.6	73.4	71.0	66.3	58.2	47.2	34.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	50.4	64.4	65.9	75.1	79.0	81.6	78.6	76.4	69.7	60.1	53.7
		MCDB	49.4	55.1	73.4	76.5	83.3	87.8	89.6	85.5	83.5	73.8	67.2	56.2
	2%	WB	38.4	44.7	59.5	61.6	70.9	76.4	78.3	76.2	73.4	65.2	55.9	43.3
		MCDB	42.3	48.2	64.1	70.5	79.0	84.5	85.7	82.6	81.1	69.5	61.6	45.5
	5%	WB	35.6	39.2	52.4	58.2	67.6	74.1	76.4	74.4	71.2	61.4	50.9	38.8
		MCDB	38.1	42.9	58.8	66.7	75.2	82.0	84.4	80.3	77.6	68.2	56.5	41.8
	10%	WB	33.2	35.5	46.1	54.5	64.5	71.6	74.6	72.4	68.5	58.1	46.7	35.8
		MCDB	35.0	38.3	50.9	63.0	71.4	80.0	82.4	78.5	74.2	65.6	51.6	38.0
Mean Daily Temperature Range	5% DB	MDBR	14.0	14.5	15.2	18.2	18.4	17.6	17.6	17.6	19.5	17.8	14.7	12.7
		MCDBR	15.5	16.3	22.5	26.4	23.4	20.2	18.8	18.8	22.3	23.4	21.1	15.3
		MCWBR	12.2	13.1	14.5	14.6	12.0	11.3	10.0	10.6	12.0	13.3	14.5	12.8
	5% WB	MCDBR	14.5	16.2	21.5	22.4	19.9	18.5	16.9	16.9	17.9	19.5	19.5	14.8
		MCWBR	12.0	13.2	14.7	14.7	12.6	12.2	10.6	10.9	11.0	14.4	15.2	12.9
Clear-Sky Solar Irradiance	taub	0.267	0.291	0.319	0.365	0.394	0.407	0.423	0.404	0.374	0.327	0.291	0.269	
	taud	2.391	2.331	2.391	2.310	2.287	2.285	2.239	2.303	2.360	2.486	2.506	2.435	
	Ebn at Noon	279	288	293	285	279	274	268	270	270	272	268	267	
	Edh at Noon	24	31	33	39	41	41	43	39	34	26	22	21	
All-Sky Solar Radiation	RadAvg	567	895	1204	1449	1718	1909	1960	1679	1368	882	587	430	
	RadStd	56	92	134	134	127	149	100	115	103	119	48	40	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (46 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page